भारतीय प्रौद्योगिकी संस्थान जोधपुर
एनएच 65, सुरपुरा बाईपास रोड, करवड़, राजस्थान 342037

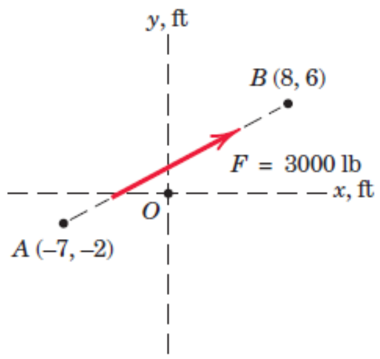
Indian Institute of Technology Jodhpur
NH 65, Surpura Bypass Rd, Karwar, Rajasthan 342037

Course: Engineering Mechanics **Code: MEL1010**

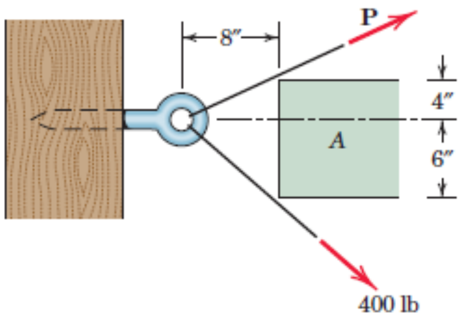
Date: 08/08/2023 **Tutorial: 01**

Question 1: If a man weighs 155 lb on earth, specify **(a)** his mass in slugs, **(b)** his mass in kilograms, and **(c)** his weight in newton. If the man is on the moon, where the acceleration due to gravity is Km = 5.30 ft/s^2, determine **(d)** his weight in pounds, and **(e)** his mass in kilograms.

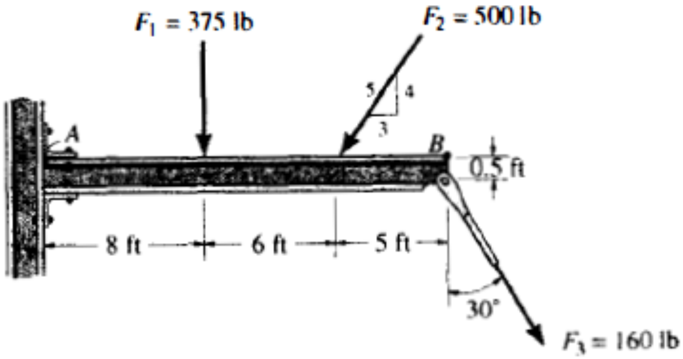
Question 2: The line of action of the 3000-lb force runs through the points *A* and *B* as shown in the figure. Determine the *x* and *y* scalar components of **F**.



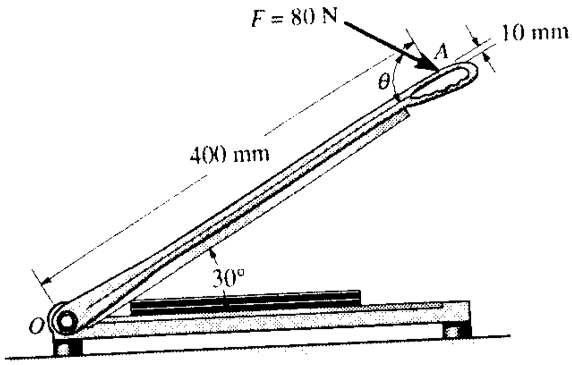
Question 3: It is desired to remove the spike from the timber by applying force along its horizontal axis. An obstruction A prevents direct access, so that two forces, one 400 lb and the other P, are applied by cables as shown. Compute the magnitude of P necessary to ensure a resultant T directed along the spike. Also find T.



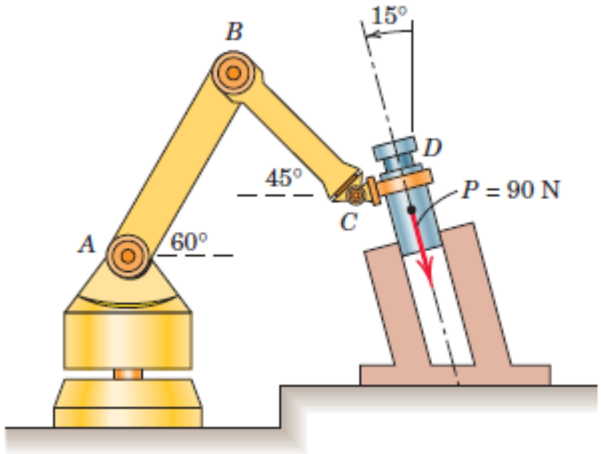
Question 4: Determine the moment about point A of each of the three forces acting on the beam.



Question 5: A force of 80 N acts on the handle of the paper cutter at A. Determine the moment created by this force about the hinge at O, if θ=60°. At what angle θ should the force be applied so that the moment it creates about point O is maximum (clockwise)? What is this maximum moment?



Question 6: In the design of the robot to insert the small cylindrical part into a close-fitting circular hole, the robot arm must exert a 90-N force P on the part parallel to the axis of the hole as shown. Determine the components of the force which the part exerts on the robot along axes (a) parallel and perpendicular to the arm AB, and (b) parallel and perpendicular to the arm BC.



Question 7: Watch the following video to understand the inception of Metric measuring system: https://www.youtube.com/watch?v=nRnuY1Vao0o&t=1026s

Measurement System

Question 8: Read and explore the history of measurement systems in India.

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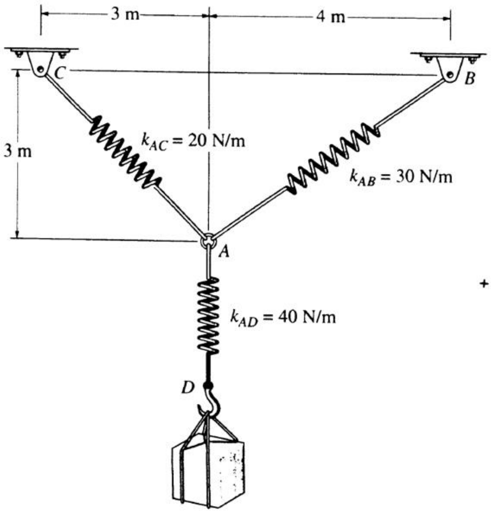
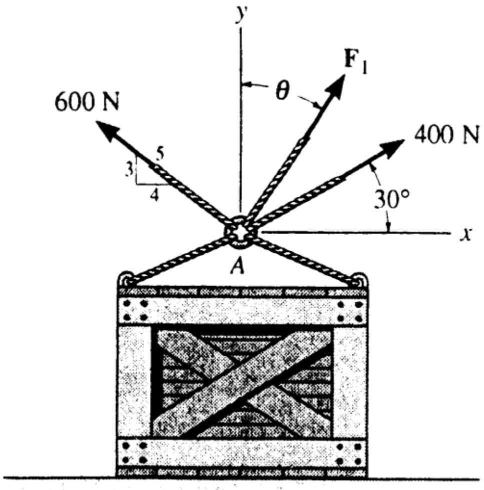
Course: Engineering Mechanics **Code: MEL1010**

Date: 23/08/2024 **Tutorial: 02**

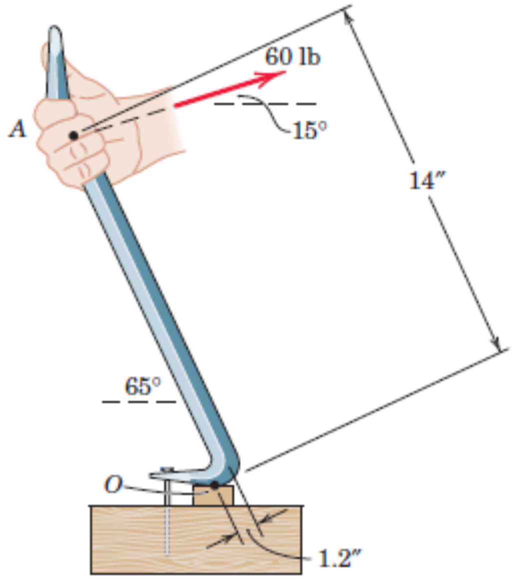
Question 01: Determine the stretch in each spring for equilibrium of 2 kg block. The springs are shown in equilibrium position.



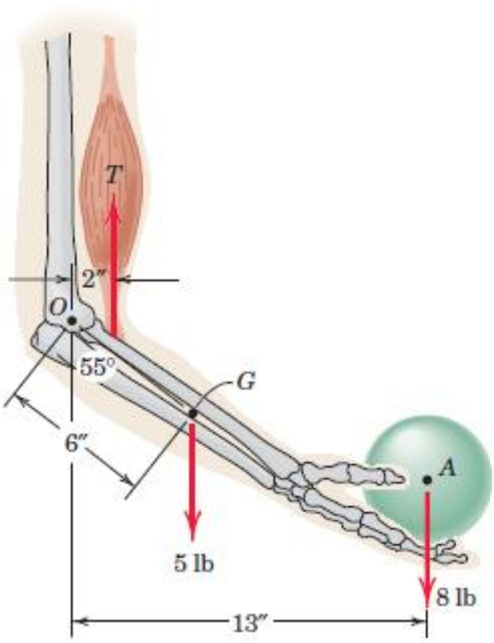
Question 02: Determine the magnitude and direction measured counterclockwise from the positive x axis of the resultant force of the three forces acting on the ring A. It is given that θ = 20 and F_1 = 500 N.



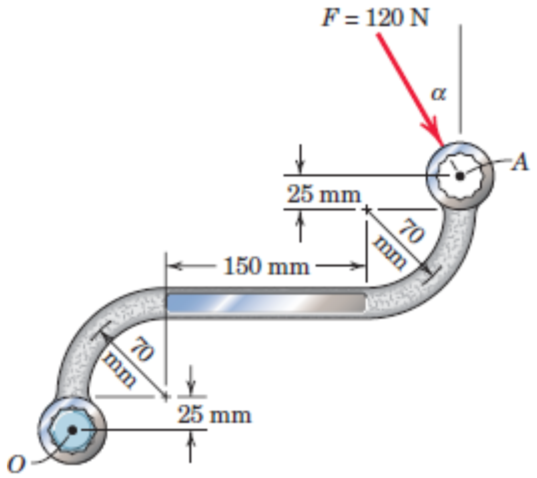
Question 03: A prybar is used to remove a nail as shown. Determine the moment of the 60-lb force about the point O of contact between the prybar and the small support block.



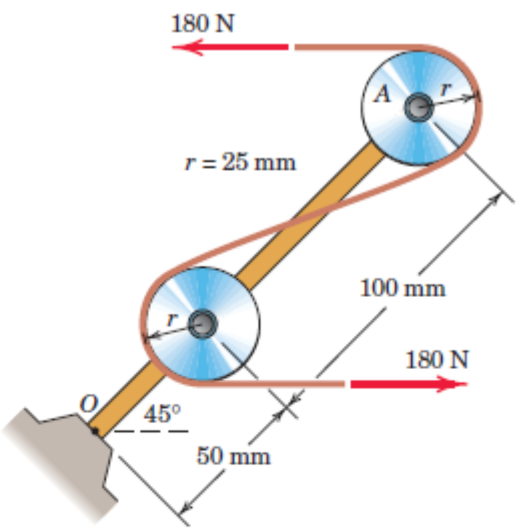
Question 04: Elements of the lower arm are shown in the figure. The weight of the forearm is 5 lb with mass center at G. Determine the combined moment about the elbow pivot O of the weights of the forearm and the sphere. What must the biceps tension force be so that the overall moment about O is zero?



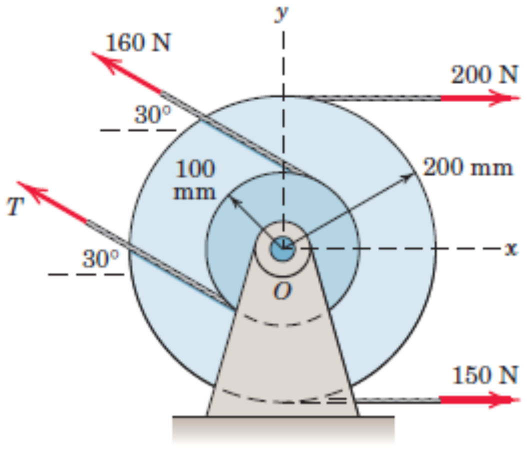
Question 05: The 120-N force is applied as shown to one end of the curved wrench. If calculate the moment of F about the center O of the bolt. Determine the value of which would maximize the moment about O; state the value of this maximum moment.



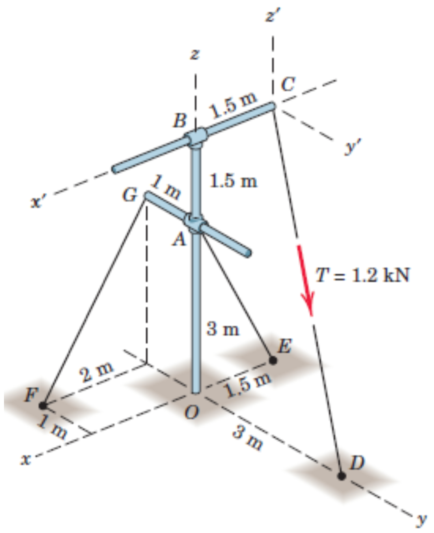
Question 06: The system consisting of the bar OA, two identical pulleys, and a section of thin tape is subjected to the two 180-N tensile forces shown in the figure. Determine the equivalent force–couple system at point O.



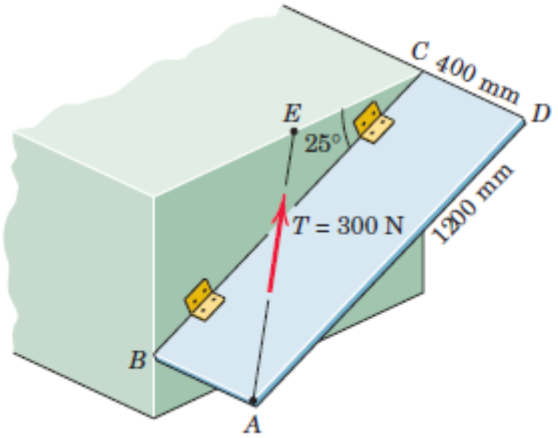
Question 07: Two integral pulleys are subjected to the belt tensions shown. If the resultant R of these forces passes through the center O, determine T and the magnitude of R and the counterclockwise angle it makes with the x-axis.



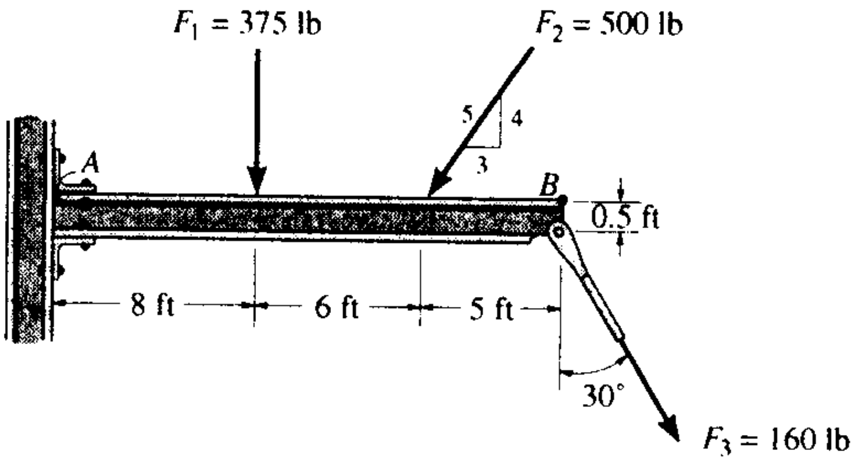
Question 08: The rigid pole and cross-arm assembly is supported by the three cables shown. A turnbuckle at D is tightened until it induces a tension T in CD of 1.2 kN. Express T as a vector. Does it make any difference in the result which coordinate system is used?



Question 09: The rectangular plate is supported by hinges along its side BC and by the cable AE. If the cable tension is 300 N, determine the projection onto line BC of the force exerted on the plate by the cable. Note that E is the midpoint of the horizontal upper edge of the structural support.



Question 10: Determine the moment about point A and B of each of the three forces acting on the beam.



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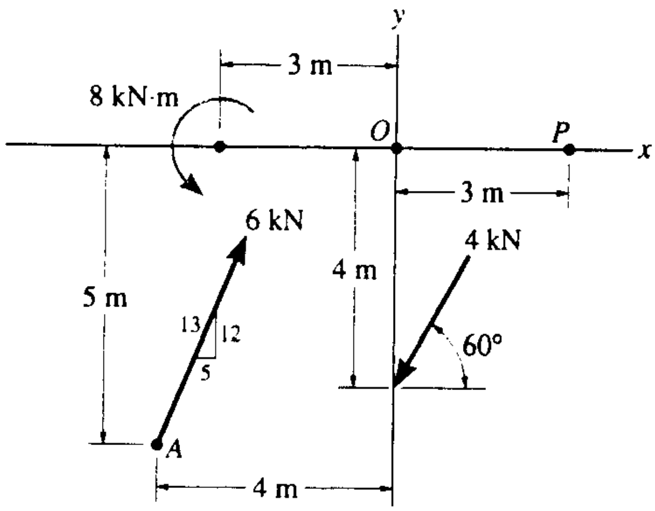
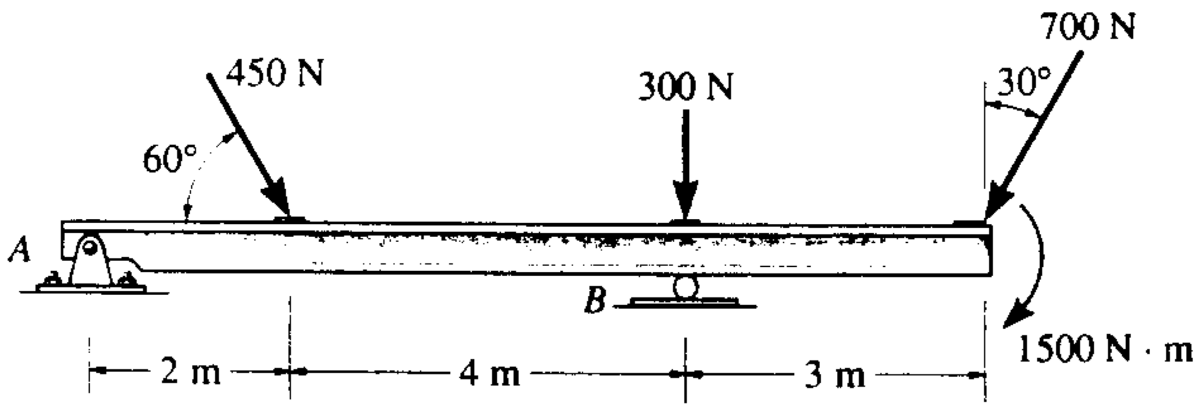
Course: Engineering Mechanics **Code: MEL1010**

Date: 29/08/2024 **Tutorial: 03**

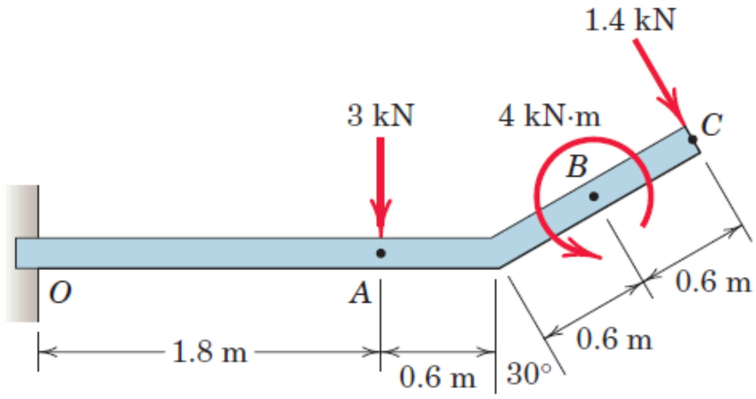
Question 01: Replace the force and couple system by an equivalent force and couple moment at point O.



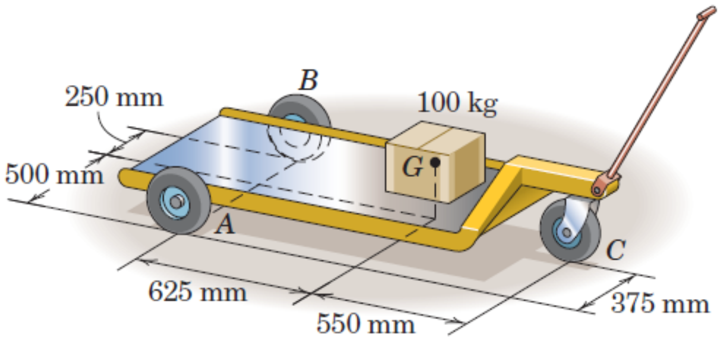
Question 02: Replace the three forces acting on the beam by a single resultant force. Specify where the force acts, Measured from end A.



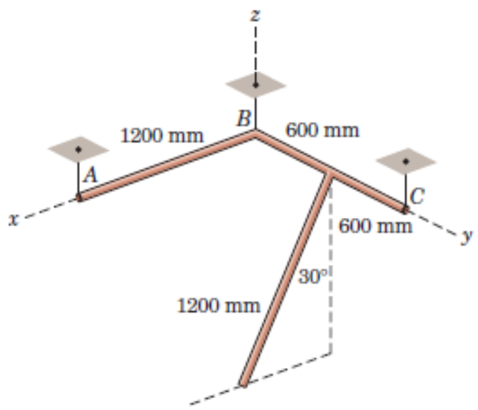
Question 03: The uniform beam has a mass of 50 kg per meter of length. Compute the reactions at the support O. The force loads shown lie in a vertical plane.



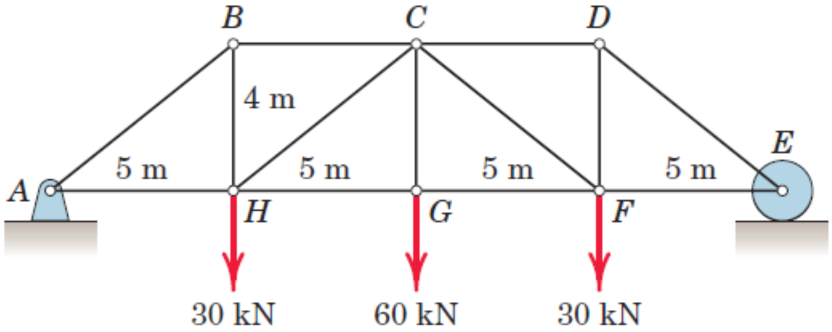
Question 04: The three-wheel truck is used to carry the 100-kg box as shown. Calculate the changes in the normal force reactions at the three wheels due to the weight of the box.



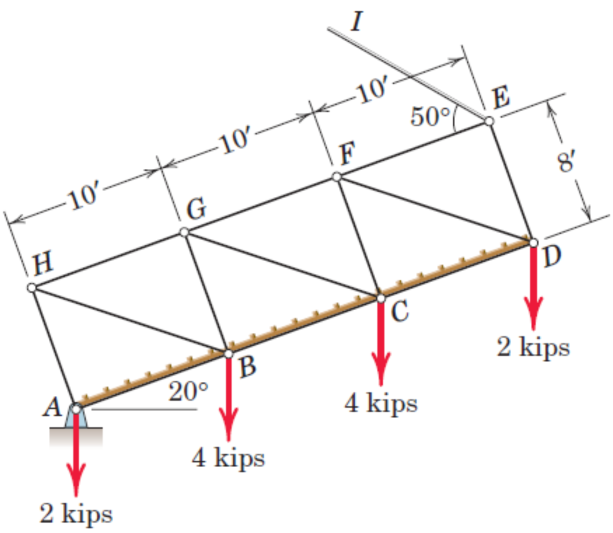
Question 05: Each of the three uniform 1200-mm bars has a mass of 20 kg. The bars are welded together into the configuration shown and suspended by three vertical wires. Bars AB and BC lie in the horizontal x-y plane, and the third bar lies in a plane parallel to the x-z plane. Compute the tension in each wire.



Question 06: Determine the force in each member of the loaded truss. Make use of the symmetry of the truss and of the loading.



Question 07: A drawbridge is being raised by a cable *EI*. The four joint loadings shown result from the weight of the roadway. Determine the forces in members *EF*, *DE*, *DF*, *CD*, and *FG.*



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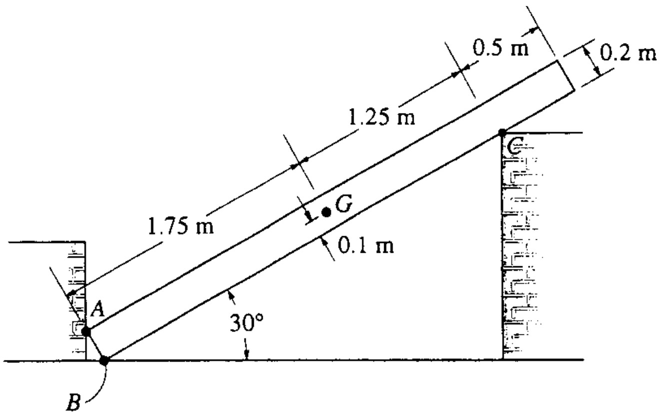
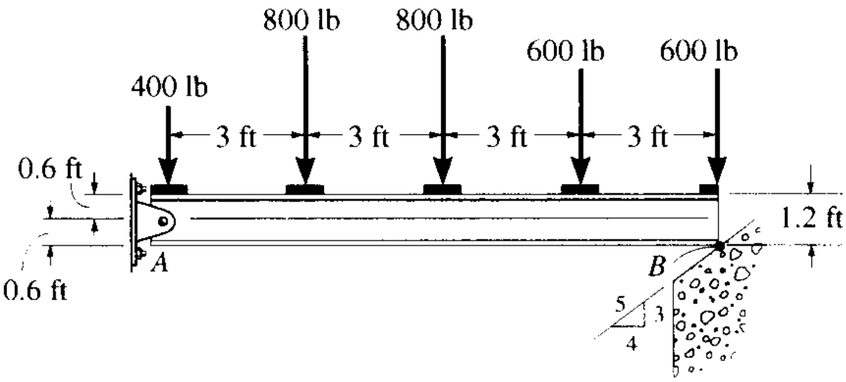
Course: Engineering Mechanics **Code: MEL1010**

Date: 05/09/2024 **Tutorial: 04**

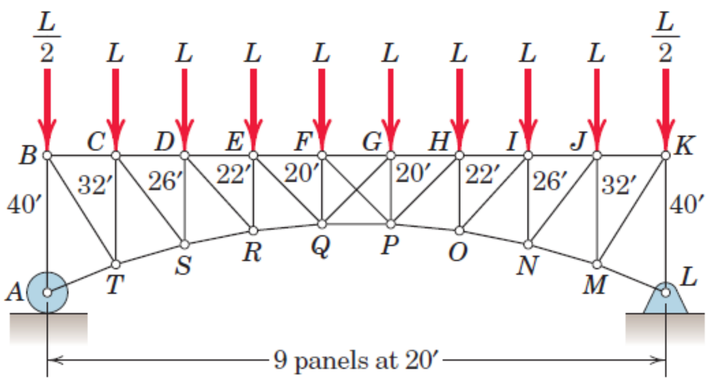
Question 01: A uniform bar has a mass of 100 kg and centre of mass at G. The supports A, B and C are smooth. Determine the reactions at the points of contact at A, B and C of the bar.



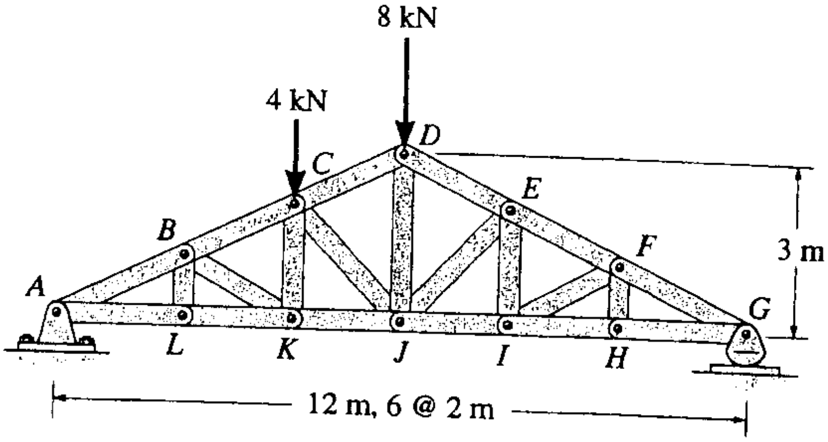
Question 02: Determine the support reactions in the following problem.



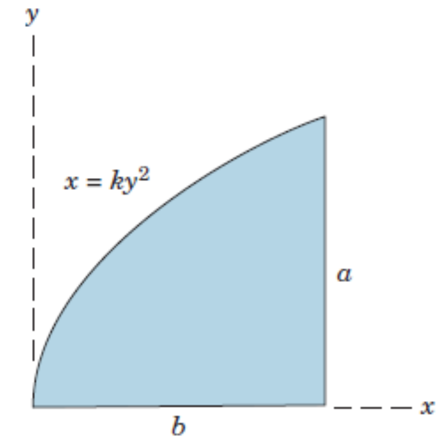
Question 03: Determine the force in member *HP* of the loaded truss. Members *FP* and *GQ* cross without touching and are incapable of supporting compression.



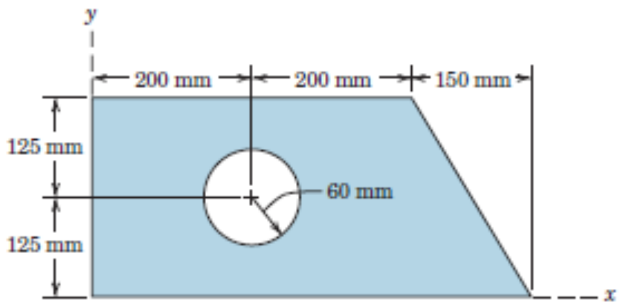
Question 04: The roof truss supports the vertical loading shown. Determine the force in members BC, CK and KJ and state if the members are in tension or compression.



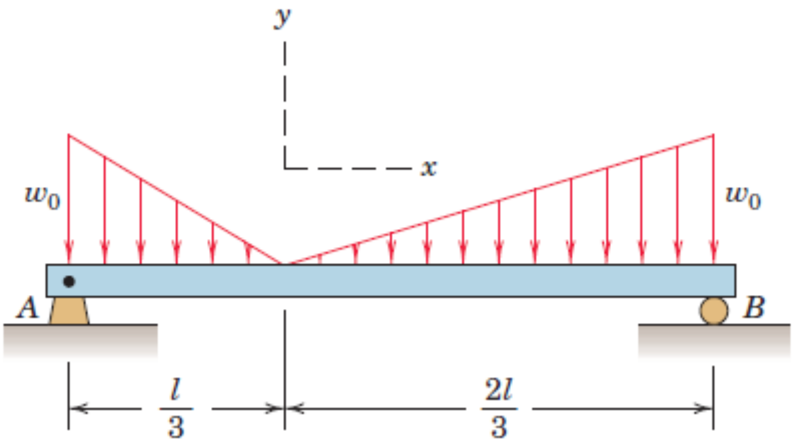
Question 05: Determine the coordinates of the centroid of the shaded area.



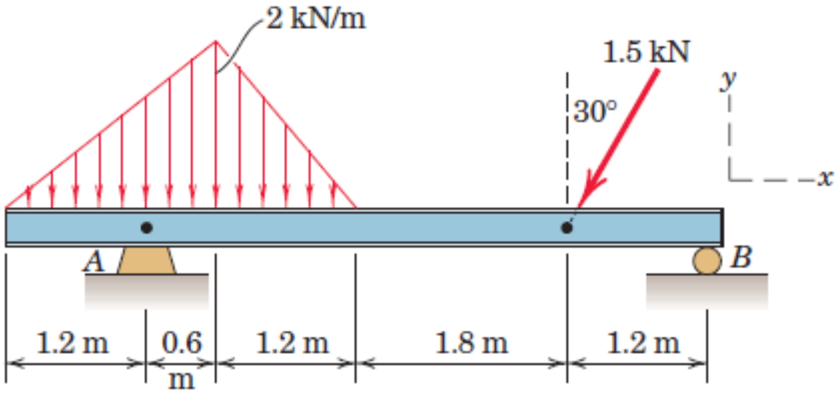
Question 06: Determine the coordinates of the centroid of the shaded area.



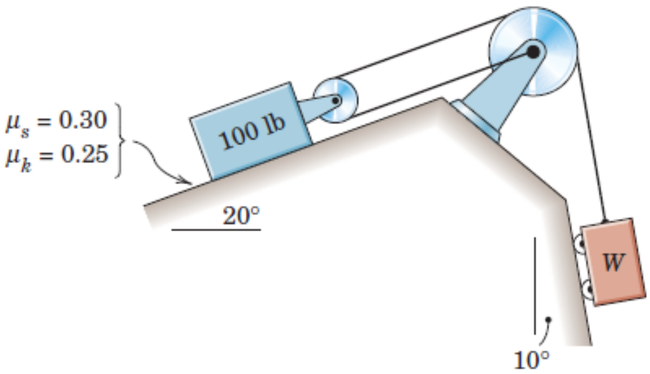
Question 07: Calculate the support reactions at A and B for the loaded beam.



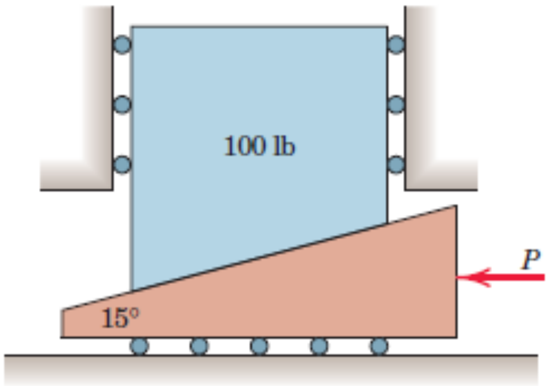
Question 08: Determine the reactions at A and B for the beam subjected to a combination of distributed and point loads.



Question 09: Determine the range of weights W for which the 100-lb block is in equilibrium. All wheels and pulleys have negligible friction.



Question 10: The coefficient of static friction μs between the 100-lb body and the 15o wedge is 0.20. Determine the magnitude of the force P required to begin raising the 100-lb body if (a) rollers of negligible friction are present under the wedge, as illustrated, and (b) the rollers are removed and the coefficient of static friction μs = 0.20 applies at this surface as well.

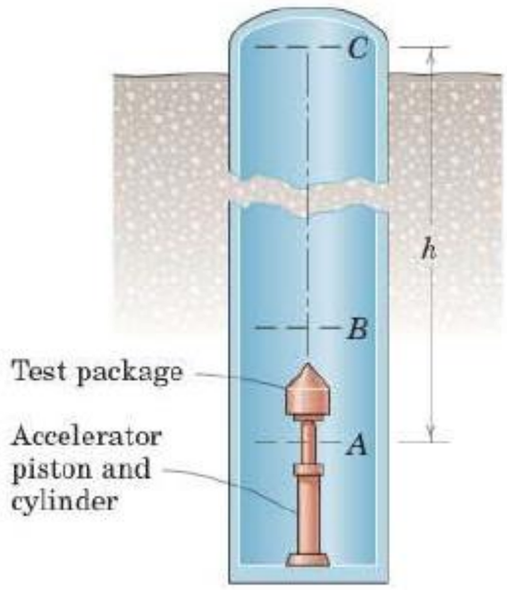


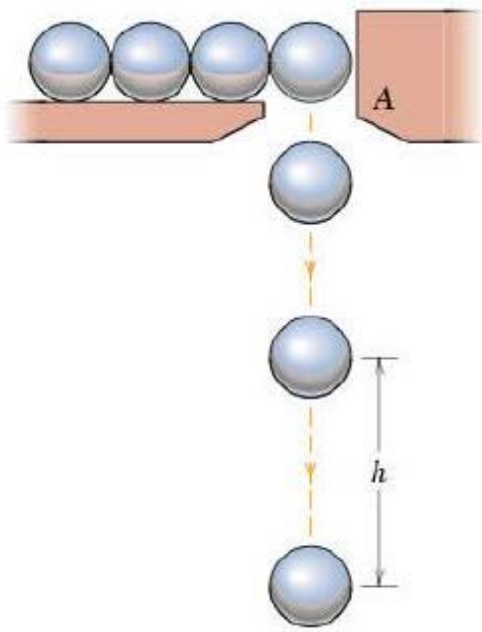
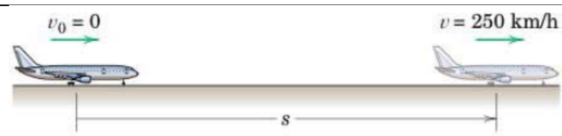
MEL1010 Engineering Mechanics
Tutorial 5 Kinematics of Particles

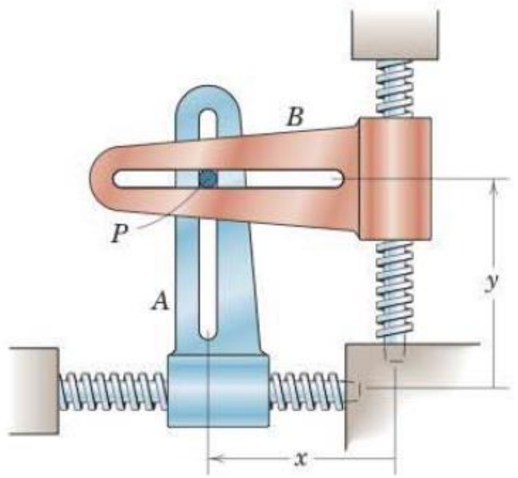
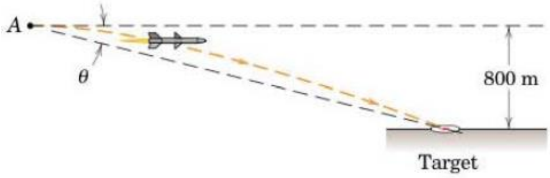
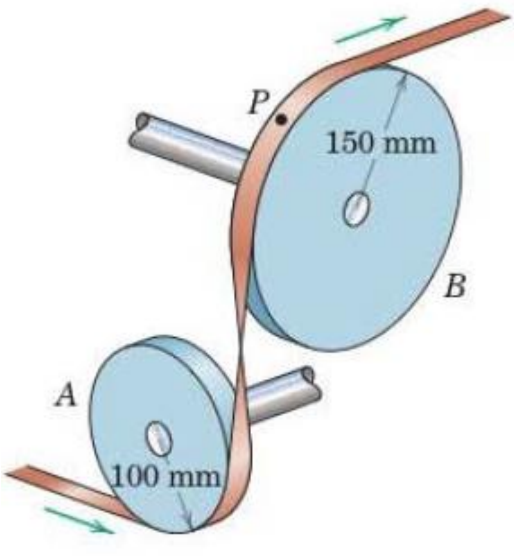
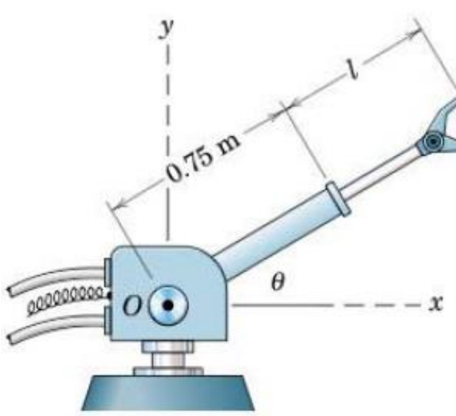
Tutorial Class: Thursday, 26th September 2024

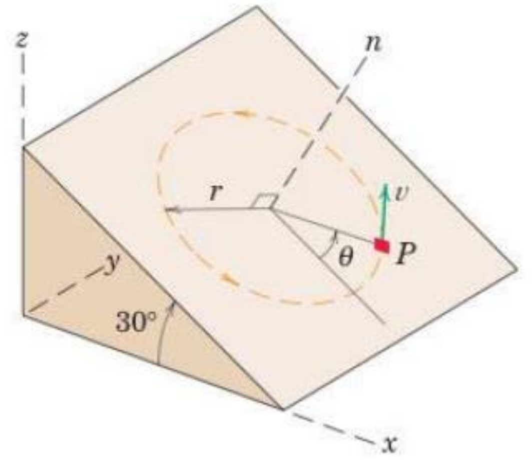
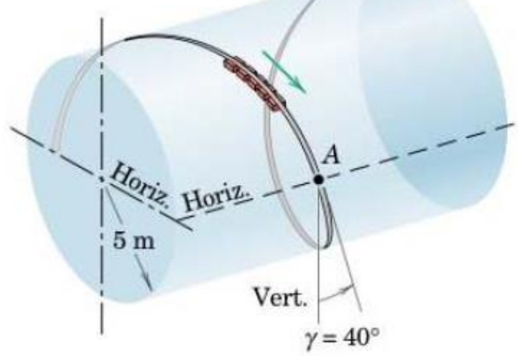
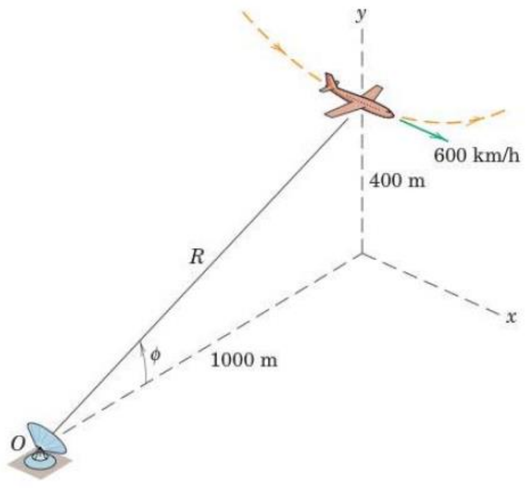
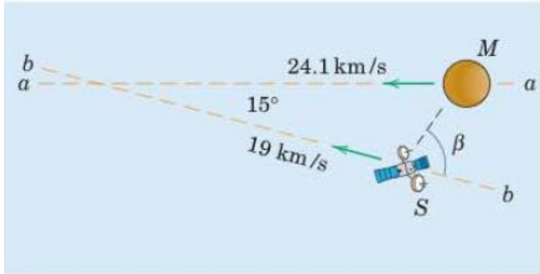
Answer following questions. Mention valid assumptions wherever applicable

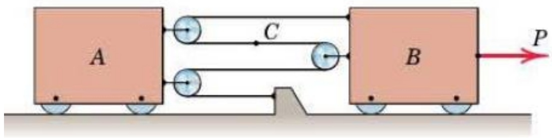
Questions about rectilinear motion, curvilinear motion using normal-tangential coordinates and polar coordinates, relative motion, space curvilinear motion, constrained motion

<table>
<tr><td>1</td><td>To test the effects of “weightlessness" for short periods of time, a test facility is designed which accelerates a test package vertically up from A to B by means of a gas-activated piston and allows it to ascend and descend from B to C to B under free-fall conditions. The test chamber consists of a deep well and is evacuated to eliminate any appreciable air resistance. If a constant acceleration of $40g$ from A to B is provided by the piston and if the total test time for the 'weightless" condition from B to C to B is 10 s, calculate the required working height h of the chamber. Upon returning to B, the test package is recovered in a basket filled with polystyrene pellets inserted in the line of fall.
$Ans.\ h = 125.7\ m$</td><td>

“Zero-g” test facility</td></tr>
<tr><td>2</td><td>Small steel balls fall from rest through the opening at A at the steady rate of two per second. Find the vertical separation h of two consecutive balls when the lower one has dropped 3 meters. Neglect air resistance.
$Ans.\ h = 2.61\ m$</td><td>
</td></tr>
<tr><td>3</td><td>On its takeoff roll, the airplane starts from rest and accelerates according to $a = a_0 - kv^2$, where a_0 is the constant acceleration resulting from the engine thrust and $-kv^2$ is the acceleration due to aerodynamic drag. If $a_0 = 2\ m/s^2$, $k =$</td><td>
</td></tr>
</table>

	$0.00004\, m^{-1}$, and v is in meters per second, determine the design length of runway required for the airplane to reach the take-off speed of $250\ km/h$ if the drag term is (a) excluded and (b) included. $Ans.\ (a)\ s\ =\ 1206\, m, (b)\ s\ =\ 1268\, m,$	
4	The x- and y- motion of guides A and B with right-angle slots control the curvilinear motion of the connecting pin P, which slides in both slots. For a short interval, the motions are governed by $x = 20 + \frac{1}{4}t^2$ and $y = 15 - \frac{1}{6}t^3$, where x and y are in millimeters and t is in seconds. Calculate the magnitude of the velocity $\boldsymbol{v}$ and acceleration $\boldsymbol{a}$ of the pin for $t = 2\ s$. Sketch the direction of the path and indicate its curvature for this instant. $Ans.\ \ v = 2.24\, mm/s,\ a\ =\ 2.06\, mm/s^2$ Note: Use R*ectangular* Coordinates	
5	A rocket is released at point A from jet aircraft flying horizontally at 1000 km/h at an altitude of 800 m. If the rocket thrust remains horizontal and gives the rocket a horizontal acceleration of $0.5g$, determine the angle θ from the horizontal to the line of sight to the target. $Ans.\quad \theta = 30°$ Note: Use R*ectangular* Coordinates	
6	The direction of motion of a flat tape in a numerical-control device is changed by the two pulleys A and B shown. If the speed of the tape increases uniformly from 2 m/s to 18 m/s while 8 meters of tape pass over the pulleys, calculate the magnitude of the acceleration of point P on the tape in contact with pulley B at the instant when the tape speed is 3 m/s. $Ans. a = 63.2\ \mathrm{m/s^2}$ Note: Use *normal-tangential* coordinate	
7	The robot arm is elevating and extending simultaneously. At a given instant, $\theta = 30°$, $\dot{\theta} = 10$ deg/s $=$ constant, $l = 0.5\, m$, $\dot{l} = 0.2\, m/s$, and $\ddot{l} = -0.3\, m/s^2$. Compute the magnitudes of the velocity $\boldsymbol{v}$ and acceleration $\boldsymbol{a}$ of the gripped part P. In addition, express $\boldsymbol{v}$ and $\boldsymbol{a}$ in terms of unit vectors $\boldsymbol{i}$ and $\boldsymbol{j}$. $Ans.\ \ v = 0.296\ \mathrm{m/s},\ a\ =\ 0.345\ \mathrm{m/s^2}$ $\boldsymbol{v}\ = 0.064\ \boldsymbol{i} + 0.289\boldsymbol{j}\ \mathrm{m/s}$ $\boldsymbol{a}\ = -0.328\boldsymbol{i} - 0.1086\boldsymbol{j}\ \mathrm{m/s^2}$ Note: Use *Polar* Coordinate	

8	The small block P travels with constant speed v in the circular path of radius r on the inclined surface. If $\theta = 0$ at time $t = 0$, determine the x-, y-, and z-components of velocity and acceleration as function of time. *Ans*. $x = \frac{\sqrt{3}}{2} r \cos\frac{v}{r}t, y = r \sin\frac{v}{r}t, z = -\frac{1}{2} r \cos\frac{v}{r}t$ $\dot{x} = -\frac{\sqrt{3}}{2} v \sin\frac{v}{r}t, \dot{y} = v \cos\frac{v}{r}t, \dot{z} = \frac{1}{2} v \sin\frac{v}{r}t$ $\ddot{x} = -\frac{\sqrt{3}}{2}\frac{v^2}{r}\cos\frac{v}{r}t, \ddot{y} = -\frac{v^2}{r}\sin\frac{v}{r}t, \ddot{z} = \frac{1}{2}\frac{v^2}{r}\cos\frac{v}{r}t$ Note: Space Curvilinear Motion, Rectangular coordinates	
9	An amusement ride called the “corkscrew” takes the passengers through the upside-down curve of a horizontal cylindrical helix. The velocity of the cars as they pass position A is 15 m/s, and the component of their acceleration measured along the tangent to the path is $g\cos\gamma$ at this point. The effective radius of the cylindrical helix is 5 m, and the helix angle is $\gamma = 40°$. Compute the magnitude of the acceleration of the passengers as they pass position A. *Ans*. $a = 27.5\ \text{m/s}^2$ Note: Space Curvilinear Motion, cylindrical coordinates	
10	At the bottom of a vertical loop in the x-y plane at an altitude of 400 m, the airplane has a speed of 600 km/h with no horizontal x-acceleration. The radius of curvature of the loop at the bottom is 1200 m. For the radar tracking at O, determine the recorded value of $\ddot{R}$ and $\ddot{\phi}$ for this instant. $Ans. \ddot{R} = 34.4\ m/s^2, \ddot{\phi} = 0.01038\ rad/s^2$ Note: Space Curvilinear Motion, Spherical Coordinates	
11	The spacecraft S approaches the planet Mars along a trajectory $b - b$ in the orbital plane of Mars with an absolute velocity of 19 km/s. Mars has a velocity of 24.1 km/s along its trajectory $a - a$. Determine the angle β between the line of sight $S - M$ and the trajectory $b - b$ when Mars appears from the spacecraft to be approaching it head on. $Ans. \beta = 55.6°$ Note: Relative Motion	

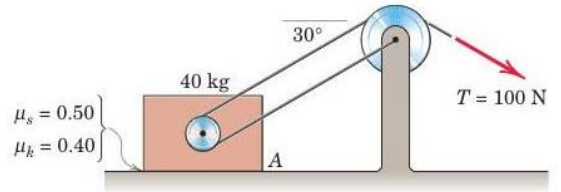
	Optional	
12	Under the action of force P, the constant acceleration of block B is 3 $\mathrm{m/s^2}$ to the right. At the instant when the velocity of B is 2 m/s to the right, determine the velocity of B relative to A, the acceleration of B relative to A, and the absolute veleocity of pont C of the cable. $Ans.\, v_{B/A} = 0.5\ \mathrm{m/s}, a_{B/A} = 0.75\ \mathrm{m/s^2}$ $v_C = 1\ \mathrm{m/s}, all\ to\ the\ right$ Note: Constrained Motion of Connected particles	

MEL1010 Engineering Mechanics
Tutorial 6 Work-Energy, Impulse Momentum and Impact

Answer all the following questions. Mention valid assumptions wherever applicable
(Note: 1-3 Newton's law, 4-6: Work Energy, 7-9: Impulse Momentum, 10-12: Impact)

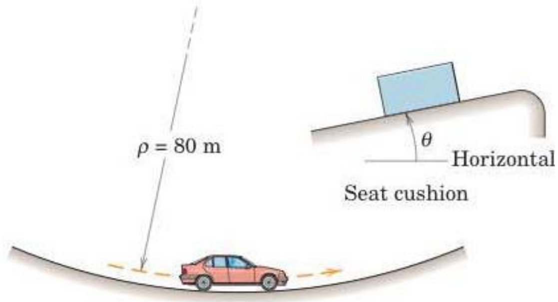
1 Compute the acceleration of block A for the instant depicted. Neglect the masses of the pulleys.

Ans. $a = 1.406\ \mathrm{m/s^2}$



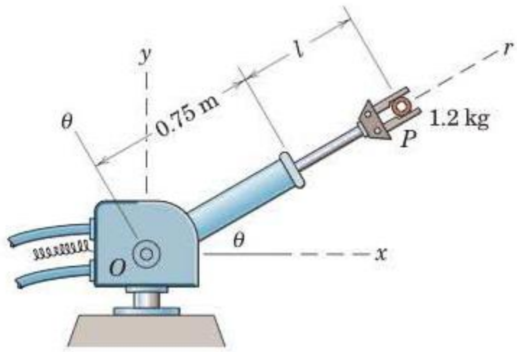
2 The car has a speed of 70 km/h at the bottom of the dip when the driver applies the brakes, causing a deceleration of $0.5g$. What is the minimum seat cushion angle θ for which a package will not slide forward? The coefficient of static friction between the package and seat cushion is (a) 0.2 and (b) 0.4.

Ans. (a) θ = 7.34°, (b) θ = -3.16°



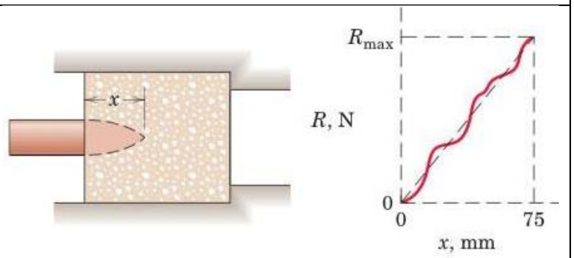
3 The robot arm is elevating and extending simultaneously. At a given instant, θ = 30°, $\dot{\theta}$ = 40 deg/s, $\ddot{\theta}$ = 120 deg/s^2, l = 0.5 m, $\dot{l}$ = 0.4 m/s, and $\ddot{l}$ = -0.3 m/s^2. Compute the radial and transverse forces F_r and F_θ that the arm must exert on the gripped part P, which has a mass of 1.2 kg. Compare with the case of static equilibrium in the same position.

Ans. F_r = 4.79 N, $(F_\theta)_{st}$= 14.00 N
$(F_r)_{st}$ = 5.89 N, $(F_\theta)_{st}$= 10.19 N



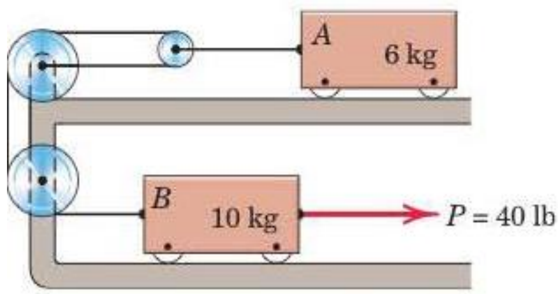
4 The resistance R to penetration x of a 0.25-kg projectile fired with a velocity of 600 m/s into a certain block of fibrous material is shown in the graph. Represent this resistance by the dashed line and compute the velocity v of the projectile for the instant when x = 25 mm if the projectile is brought to rest after a total penetration of 75 mm.

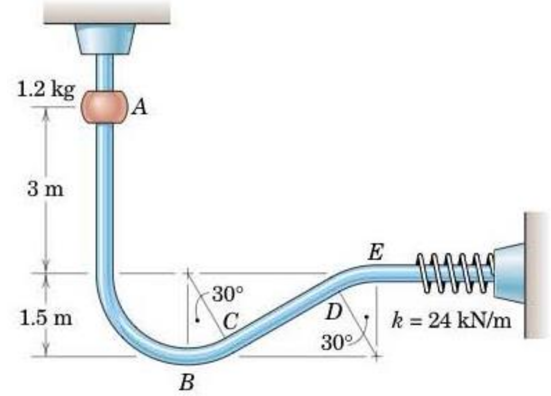
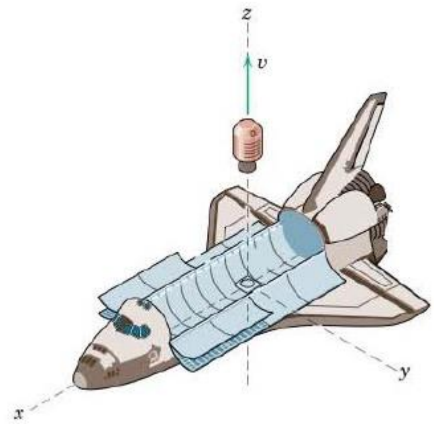
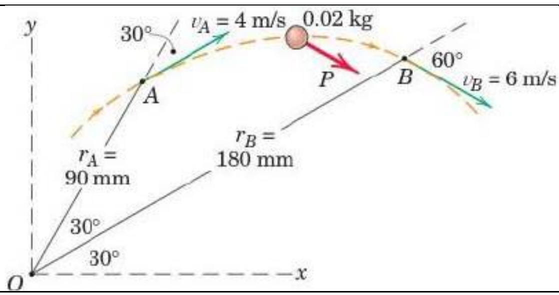
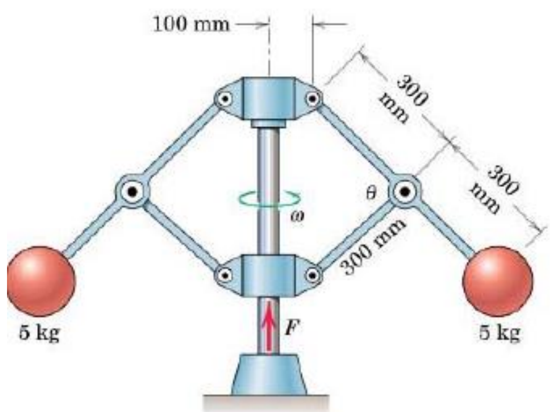
Ans. $v = 566\ \mathrm{m/s}$

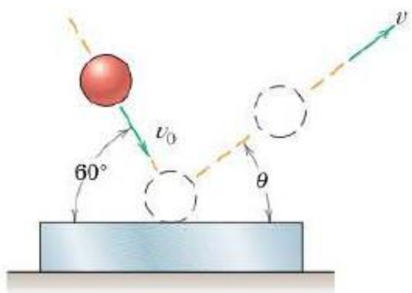
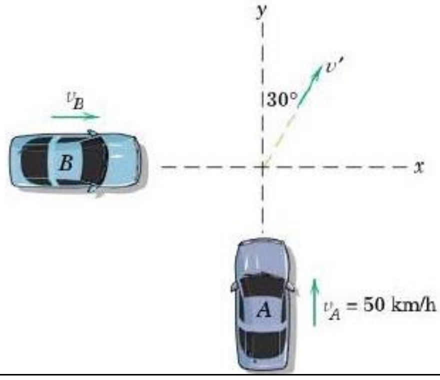
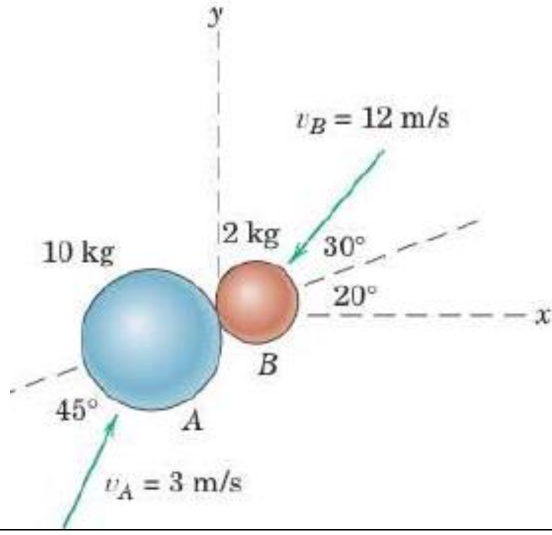


5 The force P = 40 N is applied to the system, which is initially at rest. Determine the speeds of A and B after A has moved 0.4 m.

Ans. $v_a = 1.180\ \mathrm{m/s}$, $v_b = 2.36\ \mathrm{m/s}$



6	The 1.2-kg slider is released from rest in position A and slides without friction along the vertical-plane guide shown. Determine (a) the speed v_B of the slider as it passes position B and (b) the maximum deflection δ of the spring. *Ans.* $v_b = 9.40$ m/s, $\delta = 54.2$ mm	
7	The space shuttle launches an 800-kg satellite by ejecting it from the cargo bay as shown. The ejection mechanism is activated and is in contact with the satellite for 4 s to give it a velocity of 0.3 m/s in the z-direction relative to the shuttle. The mass of the shuttle is 90 Mg. Determine the component of velocity v_f of the shuttle in the minus z-direction resulting from the ejection. Also find the time average F_{av} of the ejection force. *Ans.* $v_f = 0.00264$ m/s, $F_{av} = 59.5$ N	
8	The 0.02-kg particle moves along the dashed trajectory shown and has indicated velocities at positions A and B. Calculate the time average of the moment about O of the resultant force P acting on the particle during the 0.5 second required for it to go from A to B. *Ans.* $Mo_{av} = 0.0302$ Nm	
9	The assembly of two 5-kg spheres is rotating freely about the vertical axis at 40 rev/min with θ = 90°. If the force F which maintains the given position is increased to raise the base collar and reduce θ to 60°, determine the new angular velocity ω. Also determine the work U done by F in changing the configuration of the system. Assume that the mass of the arms and collars is negligible. *Ans.* $\omega = 3$ rad/s, $U = 5.34$ J	

10	The steel ball strikes the heavy steel plate with a velocity v_0 = 24 m/s at an angle of 60° with the horizontal. If the coefficient of restitution is e = 0.8, compute the velocity v and its direction θ with which the ball rebounds from the plate. *Ans.* $\theta = 54.2°$, $v = 20.5$ m/s	
11	The two cars collide at right angles in the intersection of two icy roads. Car A has a mass of 1200 kg and car B has a mass of 1600 kg. The cars become entangled and move off together with a common velocity v' in the direction initiated. If car A was travelling 50 km/h at the instant of impact, compute the corresponding velocity of car B just before impact. *Ans.* $v_b = 21.7$ km/h	
12	Sphere A collides with sphere B as shown in the figure. If the coefficient of restitution is e = 0.5, determine the x- and y-components of the velocity of each sphere immediately after impact. Motion is confined to the x-y plane. *Ans.* $v'_{Ax} = -1.672$ m/s, $v'_{Ay} = 1.649$ m/s $v'_{Bx} = 6.99$ m/s, $v'_{By} = -3.84$ m/s	

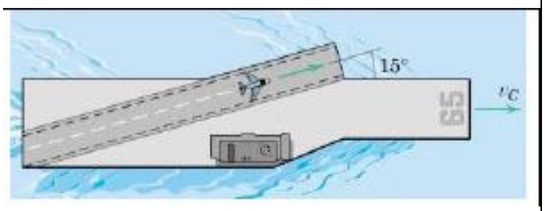
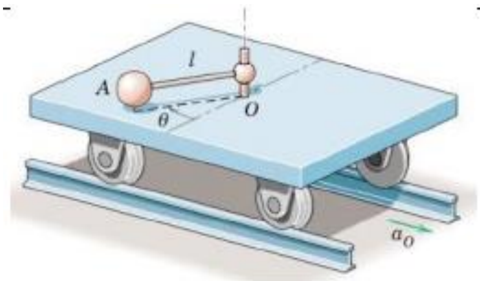
MEL1010 Engineering Mechanics

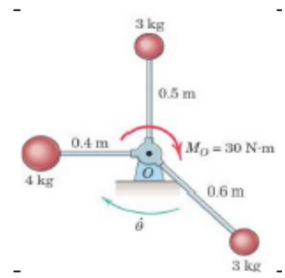
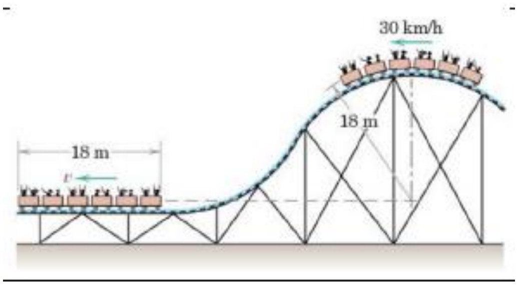
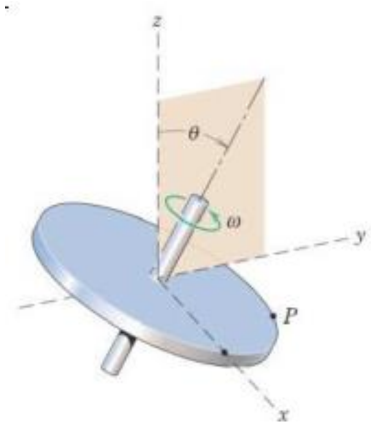
Tutorial 7: Relative Motion, System of Particle and Rigid body Kinematics-Absolute motion

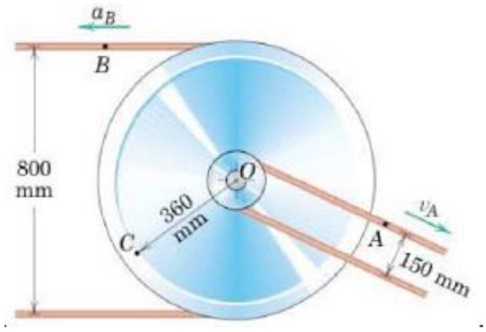
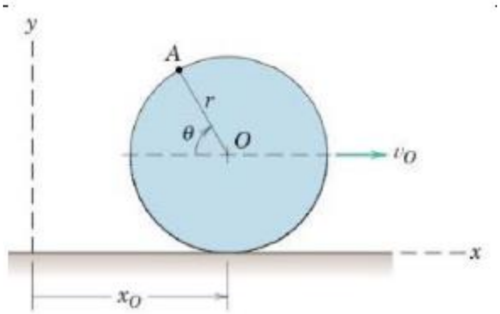
Thursday, 17th October

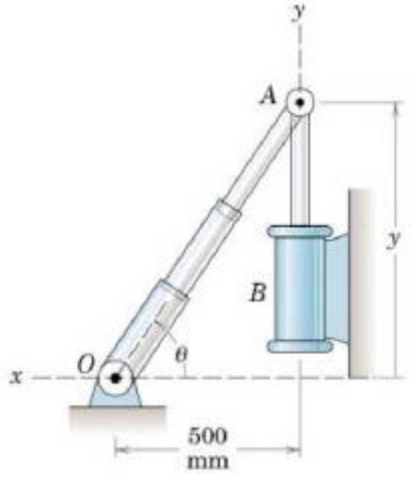
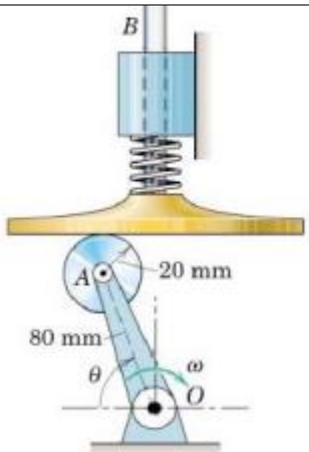
Answer all the following questions. Mention valid assumptions wherever applicable

1-2: Relative motion, 3-4: System of Particles, 5-6: Rotation, 7-9: Absolute Motion

1.	The launch catapult of the aircraft carrier gives the 7-Mg jet airplane a constant acceleration and launches the airplane in a distance of 100 m measured along the angled takeoff ramp. The carrier is moving at a steady speed $v_c = 16$ m/s. If an absolute aircraft speed of 90 m/s is desired for takeoff, determine the net force F supplied by the catapult and the aircraft engines. Ans. F= 194.0 kN	
2.	The ball A of mass 10 kg is attached to the light rod of length $l = 0.8\ m$. The mass of carriage alone is 250 kg, and it moves with an acceleration a_0 as shown. If $\dot{\theta} = 3$ rad/s when $\theta = 90°$, find the kinetic energy T of the system if the carriage has a velocity of 0.8 m/s (a) in the direction of a_0 and (b) in the direction opposite to a_0 Treat the ball as a particle. Ans. T= 112 J	

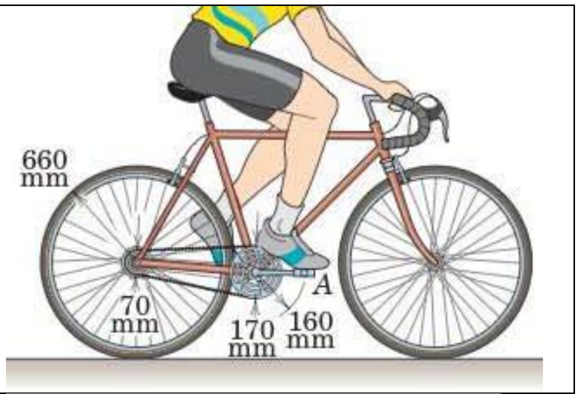
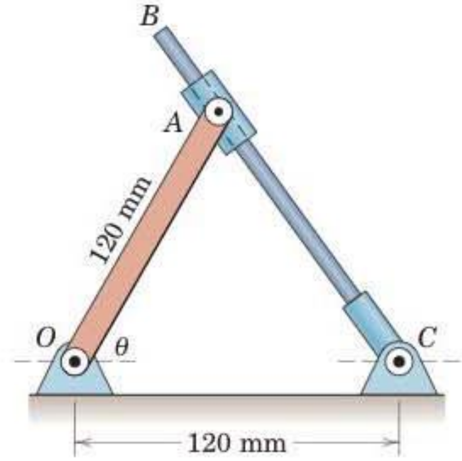
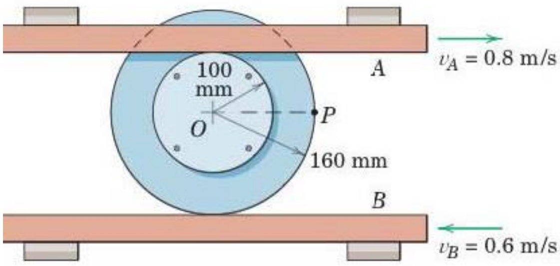
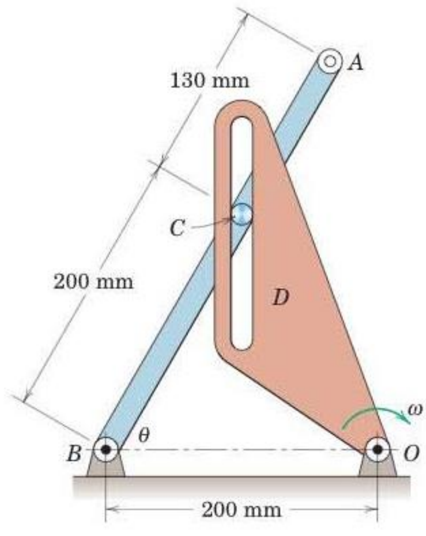
3.	The three small spheres are welded to the light rigid frame which is rotating in a horizontal plane about a vertical axis through O with an angular velocity $\dot{\theta}$ = 20 rad/s. If a couple M_0= 30 N · m is applied to the frame for 5 seconds, compute the new angular velocity $\dot{\theta}'$. *Ans.* $\dot{\theta}'$ = 80.7 rad/s	
4.	The cars of a roller-coaster ride have a speed of 30 km/h as they pass over the top of the circular track. Neglect any friction and calculate their speed v when they reach the horizontal bottom position. At the top position, the radius of the circular path of their mass centers is 18 m, and all six cars have the same mass. Ans. v = 72.7 km/h	
5.	The circular disk rotates with a constant angular velocity ω = 40 rad/sec about its axis, which is inclined in the y-z plane at the angle $\theta = \tan^{-1}\frac{3}{4}$. Determine the vector expressions for the velocity and acceleration of point P, whose position vector at the instant shown is $\boldsymbol{r} = 15\boldsymbol{i} + 16\boldsymbol{j} - 12\boldsymbol{k}$ in. (Check the magnitude of your results from the scalar values $v = r\omega$ and $a = r\omega^2$) $\boldsymbol{v} = 40(-20\,\boldsymbol{i} + 12\boldsymbol{j} - 9\boldsymbol{k})$ in/s $\boldsymbol{a} = 1600(-15\,\boldsymbol{i} - 16\boldsymbol{j} + 12\boldsymbol{k})$ in/s^2	

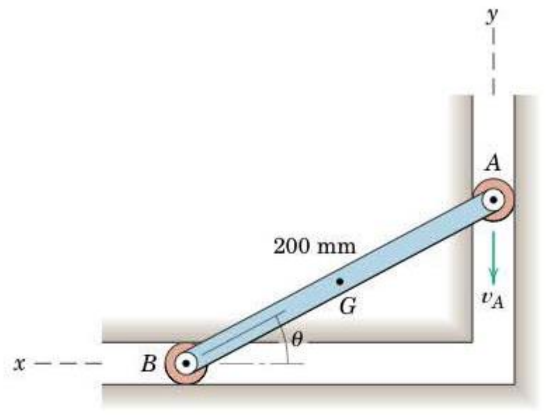
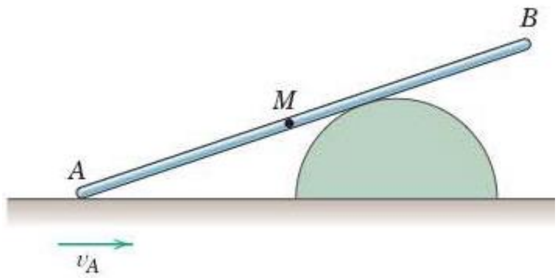
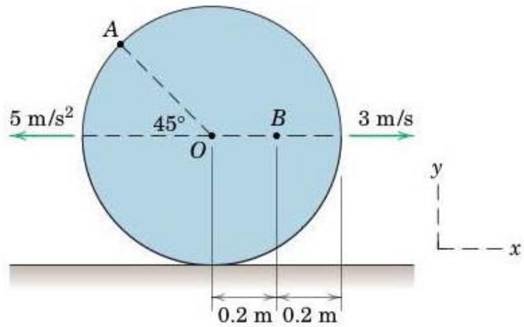
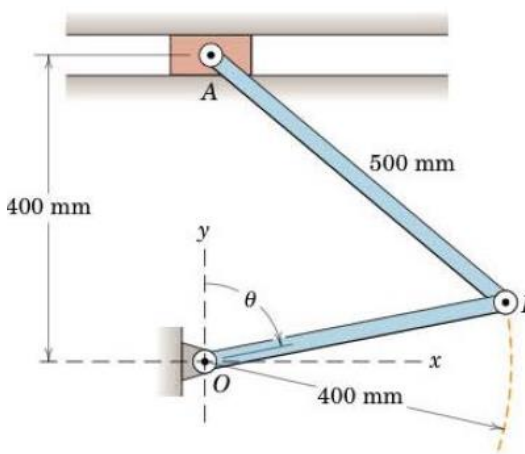
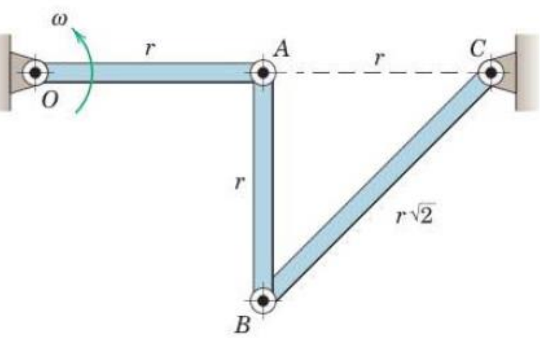
6.	The two V-belt pulleys from an integral unit and rotate about the fixed axis at O. At a certain instant, point A on the belt of the smaller pulley has a velocity v_a = 15 m/s, and point B on the belt of larger pulley has an acceleration a_B = 45 m/s^2 as shown. For this instant determine the magnitude of the acceleration a_c of point C and sketch the vector in your solution. *Ans.* a_c = 149.9 m/s^2	
7.	The wheel of radius r rolls without slipping, and its centre O has a constant velocity v_0 to the right. Determine expressions for the magnitudes of the velocity $\boldsymbol{v}$ and acceleration $\boldsymbol{a}$ of point A on the rim by differentiating its x- and y-coordinates. Represent your results graphically as vectors on your sketch and show that $\boldsymbol{v}$ is the vector sum of two vectors, each of which has a magnitude v_0 *Ans.* $v = v_0\ 2\sqrt{(1 + \sin\theta)}$, $a = \frac{v_0^2}{r}$ towards O	

8.	The telescoping link is hinged at O, and its end A is given a constant upward velocity of 200 mm/s by the piston rod of the fixed hydraulic cylinder B. Calculate the angular velocity $\dot{\theta}$ and the angular acceleration $\ddot{\theta}$ of link OA for the instant when $y = 600$ mm. *Ans.* $\dot{\theta}$ = 0.1639 rad/s $\ddot{\theta}$ = −0.0645 rad/s^2	
9.	Determine the acceleration of the shaft B for $\theta = 60°$.If the crank OA has an angular acceleration $\ddot{\theta} = 8$ rad/s^2 and an angular velocity $\dot{\theta} = 4$ rad/s at this position. The spring maintains contact between the roller and the surface of the plunger. *Ans.* a_B = 789 mm/s^2	

MEL1010 Engineering Mechanics
Tutorial 8: Relative Velocity, Instant Center of Zero Velocity and Relative Acceleration

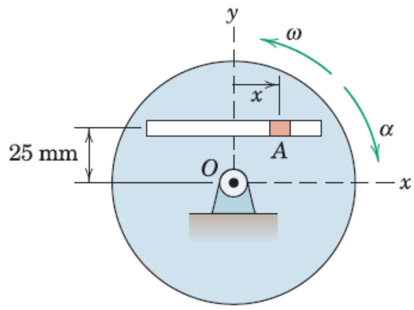
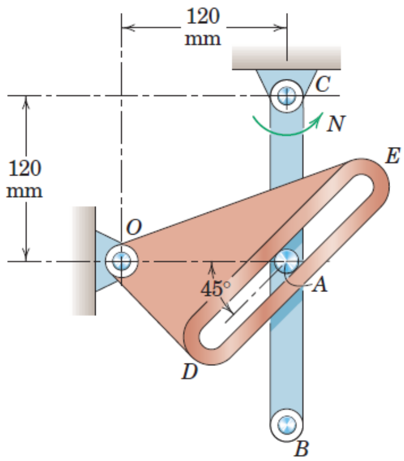
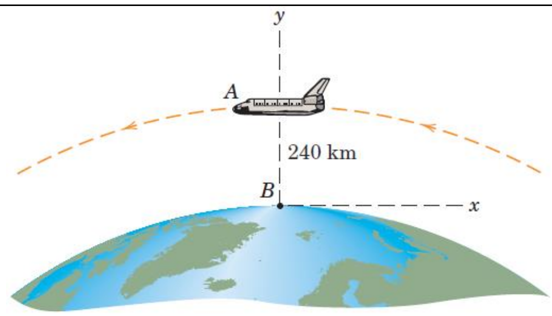
Answer all of the following questions. Mention valid assumptions wherever applicable
1-4: Relative velocity, 5-6: Instant Center of Zero Velocity, 7-9: Relative Acceleration

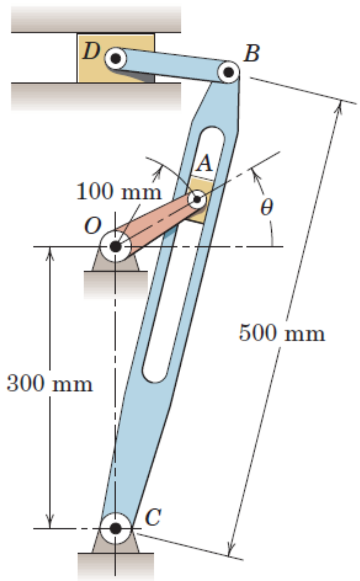
1	The rider of the bicycle shown pumps steadily to maintain a constant speed of 16 km/h against a slight head wind. Calculate the maximum and minimum magnitudes of the absolute velocity of the pedal *A*. *Ans.* $(v_A)_{max} = 5.3$ m/s,. $(v_A)_{min} = 3.56$ m/s	
2	Rod *CB* slides through the pivoted collar attached to link *OA*. If *CB* has a clockwise angular velocity of 2 rad/s, determine the angular velocity ω_{OA} of link *OA* when θ = 60°. *Ans.* $\omega_{OA} = 4$ rad/s CW	
3	Each of the sliding bars *A* and *B* engages its respective rim of the two riveted wheels without slipping. Determine the magnitude of the velocity of point *P* for the position shown. *Ans.* $v_P = 0.9$ m/s	
4	For the instant represented the rotating link *D* has an angular velocity ω = 2 rad/s, and its slot is vertical. Also θ = 60° momentarily. Determine the velocity of end *A* of link *AB* for this instant. *Ans.* $v_A = 660$ mm/s	

5	End A of the link has a downward velocity v_A of 2 m/s during an interval of its motion. For the position where θ = 30°, determine the angular velocity ω of AB and the velocity v_G of the midpoint G of the link. (Use Instant Center of Zero Velocity Method) *Ans.* $\omega = 11.55$ rad/s CW, $v_G = 1.155$ m/s	
6	End A of the slender pole is given a velocity v_A to the right along the horizontal surface. Show that the magnitude of the velocity of end B equals v_A when the midpoint M of the pole comes in contact with the semicircular obstruction. (Use Instant Center of Zero Velocity Method)	
7	The center O of the disk has the velocity and acceleration shown in the figure. If the disk rolls without slipping on the horizontal surface, determine the velocity of A and the acceleration of B for the instant represented. $\boldsymbol{v}_A = 5.12\,\boldsymbol{i} + 2.12\boldsymbol{j}$ m/s $\boldsymbol{a}_B = -16.25\,\boldsymbol{i} + 2.5\boldsymbol{j}$ m/s^2	
8	Determine the angular acceleration of link AB and the linear acceleration of A for θ = 90° if $\dot{\theta}$ = 0 and $\ddot{\theta}$ = 3 rad/s^2 at that position. Carry out your solution using vector rotation. *Ans.* $\boldsymbol{\alpha}_{AB} = -4\boldsymbol{k}$ rad/s^2, $\boldsymbol{a}_A = 1.6\,\boldsymbol{i}$ m/s^2	
9	Link OA has a constant counterclockwise angular velocity ω during a short interval of its motion. For the position shown determine the angular accelerations of AB and BC. *Ans.* $\alpha_{AB} = 2\omega^2, \alpha_{BC} = 0$	

MEL1010 Engineering Mechanics
Tutorial 9: Motion Relative to rotating axes and Kinetics using Newtonian solution approach

Answer all of the following questions. Mention valid assumptions wherever applicable

1	The disk rotates about a fixed axis through O with angular velocity $\omega = 5$ rad/s and angular acceleration $\alpha = 3\ rad/s^2$ at the instant represented, in the directions shown. The slider A moves in the straight slot. Determine the absolute velocity and acceleration of A for the same instant, when $x = 36\ mm$, $\dot{x} = -100\ mm/s$, and $\ddot{x} = 150\ mm/s^2$. Ans: $\boldsymbol{v}_A = -225\ \boldsymbol{i} + 180\boldsymbol{j}$ mm/s $\boldsymbol{a}_A = -675\ \boldsymbol{i} - 1733\boldsymbol{j}$ mm/s^2	
2	For the instant represented, link CB is rotating counterclockwise at a constant rate $N = 4$ rad/s, and its pin A causes a clockwise rotation of the slot ted member ODE. Determine the angular velocity ω and angular acceleration α of ODE for this instant. *Ans.* $\omega_{ODE} = 4$ rad/s CW, $\alpha_{ODE} = 64$rad/s^2 CCW	
3	The space shuttle A is in an equatorial circular orbit of $240 - km$ altitude and is moving from west to east. Determine the velocity and acceleration which it appears to have to an observer B fixed to and rotating with the earth at the equator as the shuttle passes overhead. Use $R = 6378$ km for the radius of the earth. Also use Fig. for the appropriate value of g and carry out your calculations to four-figure accuracy. Ans: $\boldsymbol{v}_{rel} = -26220\ \boldsymbol{i}$ km/h $\boldsymbol{a}_{rel} = -8.018\boldsymbol{j}$ m/s^2	

<table>
<tr><td>4</td><td>The figure illustrates a commonly used quick-return mechanism which produces a slow cutting stroke of the tool (attached to D) and a rapid return stroke. If the driving crank OA is turning at the constant rate $\dot{\theta} =$ 3 rad/s, determine the magnitude of the velocity of point B for the instant when $\theta =$ 30°
Ans: v_B = 288 mm/s</td><td>
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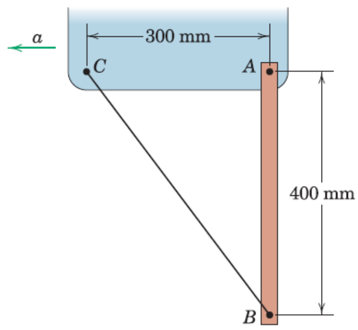
<tr><td>5</td><td>The uniform 5 kg bar AB is suspended in a vertical position from an accelerating vehicle and restrained by the wire BC. If the acceleration is a = 0.6 g, determine the tension T in the wire and the magnitude of the total force supported by the pin at A.
Ans: T = 24.5 N, A = 32.9 N</td><td>
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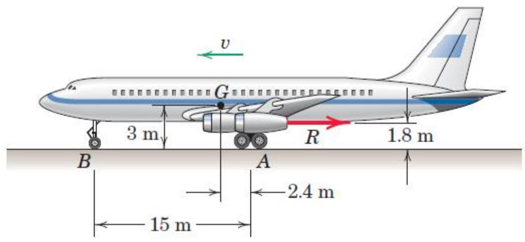
<tr><td>6</td><td>A jet transport with a landing speed of 200 km/h reduces its speed to 60 km/h with a negative thrust R from its jet thrust reversers in a distance of 425 m along the runway with constant deceleration. The total mass of the aircraft is 140 Mg with mass center at G. Compute the reaction N under the nose wheel B toward the end of the braking interval and prior to the application of mechanical braking. At the lower speed, aerodynamic forces on the aircraft are small and may be neglected.
Ans: N = 257 N</td><td>
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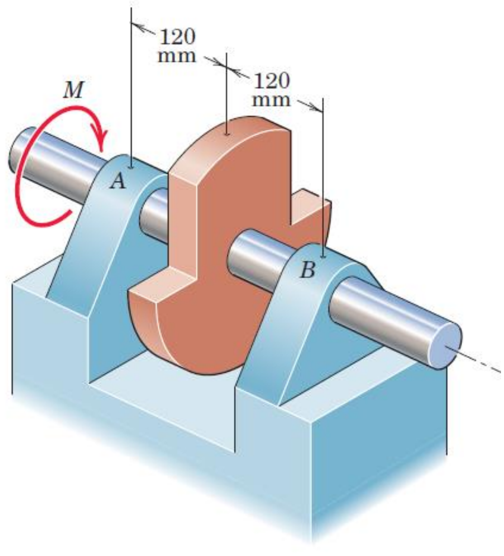
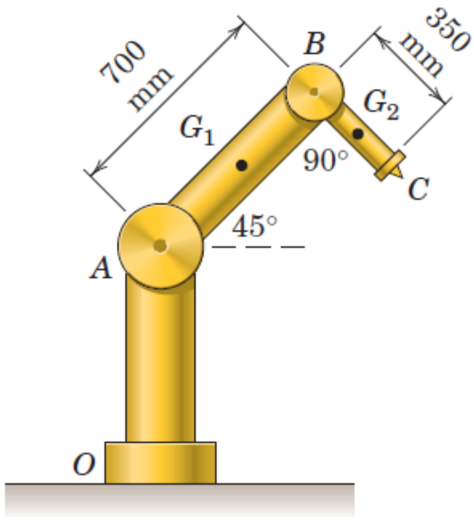
7 A vibration test is run to check the design adequacy of bearings A and B. The unbalanced rotor and attached shaft have a combined mass of 2.8 kg. To locate the mass center, a torque of 0.660 Nm is applied to the shaft to hold it in equilibrium in a position rotated 90° from that shown. A constant torque $M = 1.5\ N\ m$ is then applied to the shaft, which reaches a speed of 1200 rev/min in 18 revolutions starting from rest. During each revolution the angular acceleration varies, but its average value is the same as for constant acceleration. Determine (a) the radius of gyration k of the rotor and shaft about the rotation axis, (b) the force F which each bearing exerts on the shaft immediately after M is applied, and (c) the force R exerted by each bearing when the speed of 1200 rev/min is reached and M is removed. Neglect any frictional resistance and the bearing forces due to static equilibrium.

Ans: $k = 87.6$ mm, $F = 2.35$ N, $\mathrm{R} = 531$ N



8 The robotic device consists of the stationary pedestal OA, arm AB pivoted at A, and arm BC pivoted at B. The rotation axes are normal to the plane of the figure. Estimate (a) the moment M_A, applied to arm AB required to rotate it about joint A at 4 rad/s^2 counter clockwise from the position shown with joint B locked and (b) the moment M_B applied to arm BC required to rotate it about joint B at the same rate with joint A locked. The mass of arm AB is 25 kg and that of BC is 4 kg, with the stationary portion of joint A excluded entirely and the mass of joint B divided equally between the two arms. Assume that the centers of mass G_1 and G_2 are in the geometric centers of the arms and model the arms as slender rods.

Ans: $M_A = 108.9$ N.m, $M_B = 5.51$ N.m

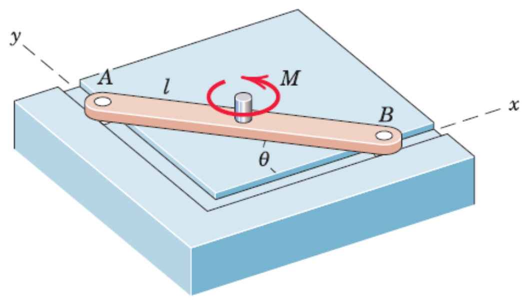


9 Small ball bearing -rollers mounted on the ends of the slender bar of mass m and length l constrain the motion of the bar in the horizontal $x - y$ slots. If a couple M is applied to the bar initially at rest at $\theta = 45^{\circ}$, determine the forces exerted on the rollers at A and B as the bar starts to move.

Ans:

$$\boldsymbol{A} = \frac{3M}{2\sqrt{2}l}\boldsymbol{i}$$

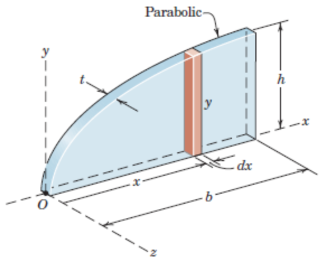
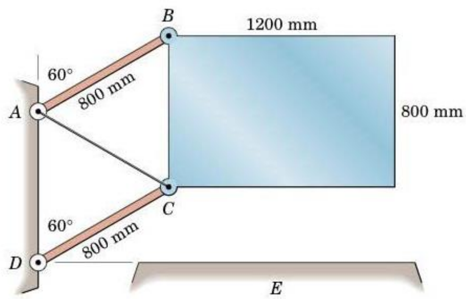
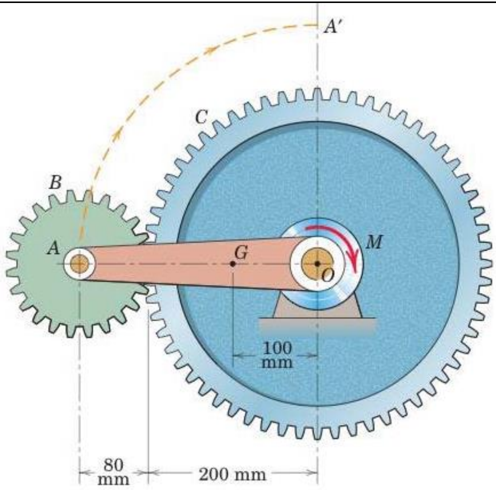
$$\boldsymbol{B} = -\frac{3M}{2\sqrt{2}l}\boldsymbol{j}$$

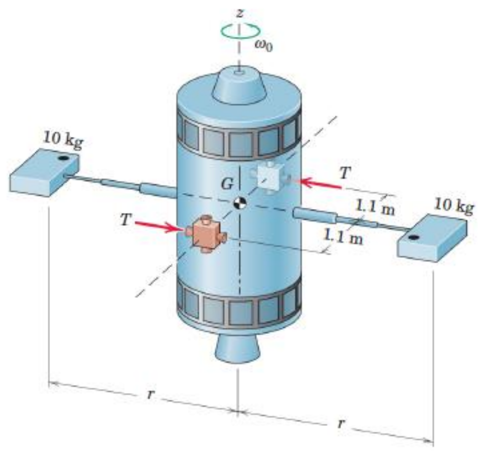


MEL1010 Engineering Mechanics

Tutorial 10 Dynamics using Work-Energy, Impulse-Momentum Approach and Rigid body Dynamics in 3 Dimensions

Answer all of the following questions. Mention valid assumptions wherever applicable

1	The upper edge of the thin homogeneous plate of mass m is parabolic with a vertical slope at the origin O. Determine its mass moments of inertia about the x-, y-, and z-axes. $Ans.\ I_{xx} = \frac{1}{5}mh^2, I_{yy} = \frac{3}{7}mb^2, Izz = m(\frac{h^2}{5} + \frac{3b^2}{7})$	
2	The uniform rectangular plate has a mass of 300 kg and is supported in the vertical plane by the two parallel links of negligible mass and by the cable AC. If the cable suddenly breaks, determine the angular velocity ω of the links an instant before the plate strikes the horizontal surface E. Also find the force in member DC at the same instant. (Use Work Energy Method) $Ans.\ \omega \quad = 3\frac{\text{rad}}{\text{s}}, F_{DC} = 1472\text{N}$	
3	For the assembly shown, arm OA has a mass of 0.8 kg and a radius of gyration about O of 140 mm. Gear B has a mass of 0.9 kg and may be treated as a solid circular disk. Gear C is fixed in the vertical plane and cannot rotate. If a constant moment $M = 4\ N.m$ is applied to arm OA, initially at rest in the horizontal position shown, calculate the velocity v of point A as it reaches the top at A'. (Use Work Energy Method) $Ans.\ v \quad = 1.976\ \text{m/s}$	

<table>
<tr><td>4</td><td>Two small variable-thrust jets are actuated to keep the spacecraft angular velocity about the z-axis constant at $\omega_0 = 1.25\ rad/s$ as the two tele- scoping booms are extended from $r_1 = 1.2\ m$ to $r_2 = 4.5\ m$ at a constant rate over a 2-min period. Determine the necessary thrust T for each jet as a function of time where $t = 0$ is the time when the telescoping action is begun. The small $10\ kg$ experiment modules at the ends of the booms may be treated as particles, and the mass of the rigid booms is negligible.
(Use Impulse Momentum Method)
Ans. $T = 0.750 + 0.01719t$</td><td>
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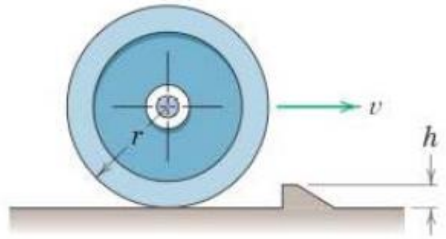
<tr><td>5</td><td>Determine the minimum velocity v which the wheel must have to just roll over the obstruction. The centroidal radius of gyration of the wheel is k, and it is assumed that the wheel does not slip.
(Use Impulse Momentum Method)
Ans. $v = \frac{r}{k^2+r^2-rh}\sqrt{(2gh(k^2+r^2))}$</td><td>
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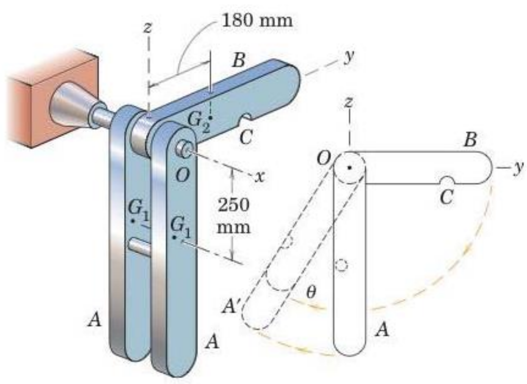
<tr><td></td><td></td><td></td></tr>
<tr><td>6</td><td>The three bars are free to rotate about the fixed horizontal x-axis at O. Each of the bars A has a mass of $20\ kg$, a radius of gyration about the x-axis of $300\ mm$, and a mass center at G_1 =. The $8\ kg$ bar B, with a radius of gyration about the same axis of $220\ mm$ and a mass center at G_2, is released from rest in the horizontal (y-axis) position and becomes attached to bars A by a latch at C at the bottom of its swing. Calculate the angle θ through which the three bars rotate as a unit and find the loss $|\Delta E|$ of energy due to the impact.
(Use Impulse Momentum Method)
Ans. $\theta = 8.97°, |\Delta E| = 12.75$ J</td><td>
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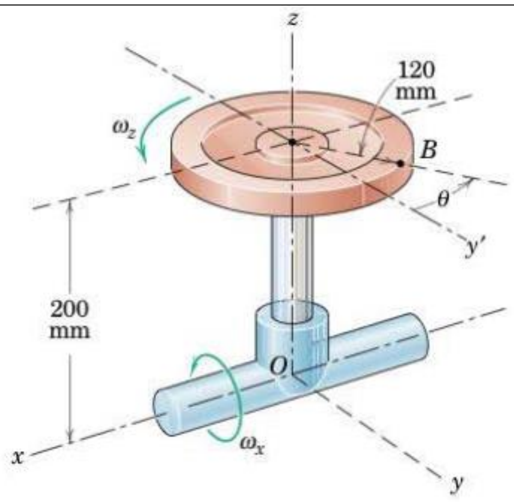
<tr><td></td><td></td><td></td></tr>
<tr><td>7</td><td>The circular disk of $120\ mm$ radius rotates about the z-axis at the constant rate $\omega_z = 20\ rad/s$, and the entire assembly rotates about the fixed x-axis at the constant rate $\omega_x = 10\ rad/s$. Calculate the magnitudes of the velocity **v** and acceleration **a** of point B for the instant when $\theta = 30°$.
Ans. $v = 3.95$ m/s, $a = 72.2$ m/s²</td><td>
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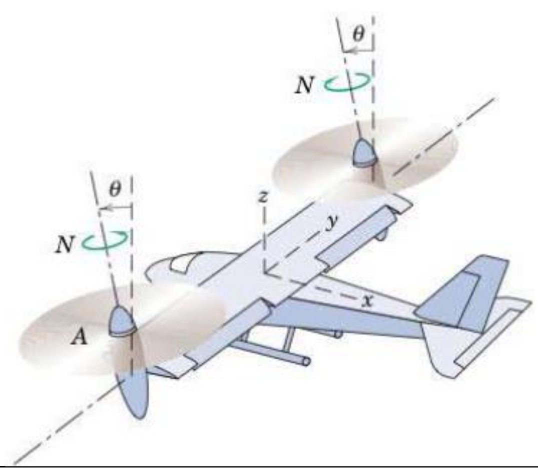
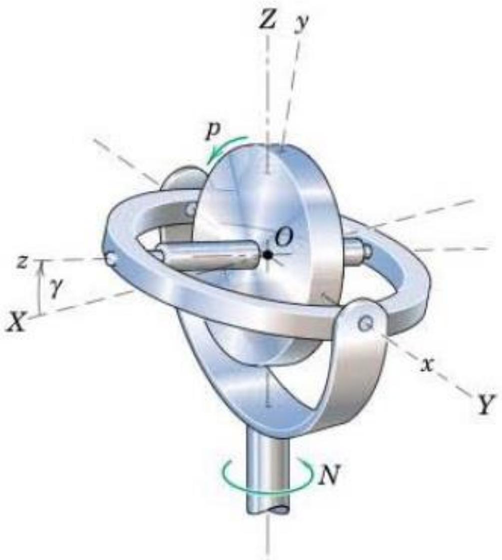
8	An unmanned radar-radio controlled aircraft with tilt-rotor propulsion is being designed for reconnaissance purposes. Vertical rise begins with $\theta = 0$ and is followed by horizontal flight as θ approaches 90°. If the rotors turn at a constant speed N of $360\, rev/min$, determine the angular acceleration α of rotor A for $\theta = 30°$ if $\dot{\theta}$ is constant at $0.2\, rad/s$. Ans: $\boldsymbol{\alpha} = -1.2\pi(\sqrt{3}\,\boldsymbol{i} + \boldsymbol{k})$ rad/s^2	
9	The gyro rotor is spinning at the constant rate $p = 100\, rev/min$ relative to the x-y-z axes in the direction indicated. If the angle γ between the gimbal ring and horizontal X-Y plane is made to increase at the rate of $4\, rad/sec$ and if the unit is forced to precess about the vertical at the constant rate $N = 20\, rev/min$, calculate the angular momentum $\boldsymbol{Ho}$ of the rotor when $\gamma = 30°$. The axial and transverse moments of inertia are $I_{zz} = 5\left(10^{-3}\right) lb\text{-}ft\text{-}sec^2$ and $I_{xx} = I_{yy} = 2.5\left(10^{-3}\right) lb\text{-}ft\text{-}sec^2$ Ans: $\boldsymbol{H}_O = -0.01\,\boldsymbol{i} + 0.0045\boldsymbol{j} + 0.0576\boldsymbol{k}$ lb-ft-sec	
10	The solid circular disk of mass $m = 2\, kg$ and radius $r = 100\, mm$ rolls in a circle of radius $b = 200\, mm$ on the horizontal plane without slipping. If the center-line OC of the axle of the wheel rotates about the z-axis with an angular velocity $\omega = 4\pi\, rad/s$, determine the expression for the angular momentum of the disk with respect to the fixed-point O. Also compute the kinetic energy of the wheel. Ans: $\boldsymbol{v}_A = 0.251(-\boldsymbol{j} + 4.25\boldsymbol{k})$N. m. s $T = 9.87$ J